PROJECT USE CASE

No.	Project use case	IoT data in ThingSpeak	ML task	Suitable ML models	ML output
1	**Smart Agriculture Crop Stress Prediction**	Temperature, humidity, soil moisture, light	Classification	Random Forest, Decision Tree,	Healthy / Moderate Stress / Severe Stress
2	**Predictive Maintenance of Motors**	Vibration, temperature, current, RPM	Classification / Regression	Decision Tree, Random Forest,	Normal / Fault; remaining useful life
3	**IoT-Based Air Quality Prediction**	PM2.5, PM10, CO_2, temperature, humidity	Regression	Linear Regression, Random Forest,	Future AQI / pollution level
4	**Smart Building Energy Consumption Prediction**	Temperature, humidity, occupancy, voltage, current	Regression	Random Forest,	Predicted energy consumption
5	**IoT-Based Water Quality Classification**	pH, turbidity, temperature, TDS	Classification	Decision Tree, Random Forest,	Safe / Unsafe water
6	**Smart Healthcare Patient Risk Monitoring**	Heart rate, SpO_2, temperature, activity	Classification	Random Forest,	Normal / At-risk
7	**IoT Network Intrusion Detection**	Packet rate, latency, bytes, connections, failed requests	Classification	Random Forest,	Normal / Attack
8	**Smart Traffic Congestion Prediction**	Vehicle count, speed, occupancy, time	Classification / Regression	Random Forest, Regression,	Congestion level / traffic volume
9	**Smart Waste-bin Fill-Level Prediction**	Ultrasonic level, weight, temperature, time	Regression / Classification	Random Forest, Linear Regression	Fill percentage / Full–Not Full

Frequently used Repositories for Datasets

1. Datasets - UCI Machine Learning Repository
2. Vyber07/cyber-security · Datasets at Hugging Face
3. vyber07 (MEGHKUMAR SHAH) · GitHub
4. Data.gov Home - Data.gov
5. Kaggle datasets